IB MATHEMATICS SL • UNIT 6.01

C Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests

PRACTICE WORKSHEET

25 Structured Exam-Style Questions | **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests Practice Exercises

1 MCQ

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$.

Two-Way Contingency Counts

	Outcome +	Outcome –
Group I	42	18
Group II	28	32

A 0.65

B 0.80

C 0.15

D 0.85

2 MCQ

The following bar chart shows the number of students who achieved different grades in a mathematics test.\nThe grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students.\nWhat is the total number of students who took the test?

A 70

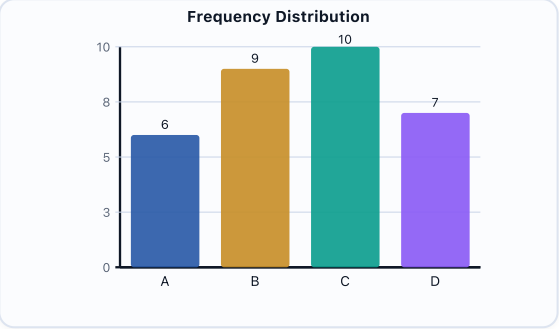
B 80

C 78

D 60

3 MCQ

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?



A 50

B 60

C 70

D 80

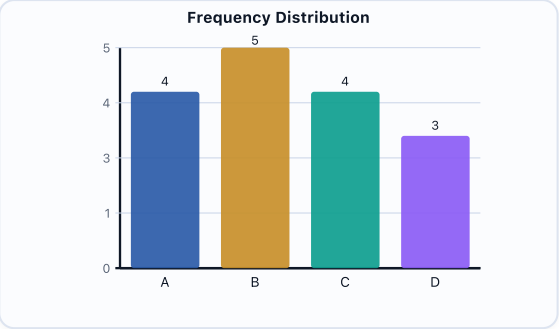
4 MCQ

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

A 20
B 25
C 30
D 35

5 MCQ

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?



A 30
B 28
C 26
D 32

6 MCQ

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

A 50
B 45
C 60
D 55

7 MCQ

Find the mean of the data set: $20, 13, 12, 12, 8$.

A 10
B 13
C 8
D 11

8 MCQ

Find the mean of the data set: $10, 1, 5, 14, 3, 7, 0, 8$.

A 6
B 4
C 3
D 1

9 MCQ

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time t is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time t. Then, determine the value of t for which the particle's speed is minimized and state the minimum speed.

A $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$

B $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed 1 at $t = 0$

C $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed 0 at $t = 0$

D $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

10 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

A $y = 2x - 5$

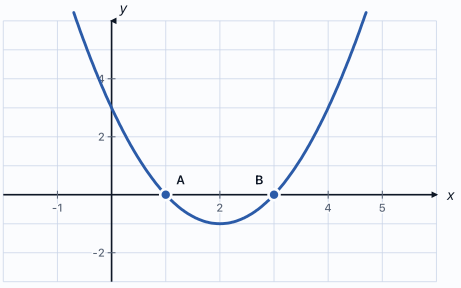
B $y = -\frac{1}{2}x + 5$

C $y = 2x - 8$

D $y = -2x + 11$

11 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

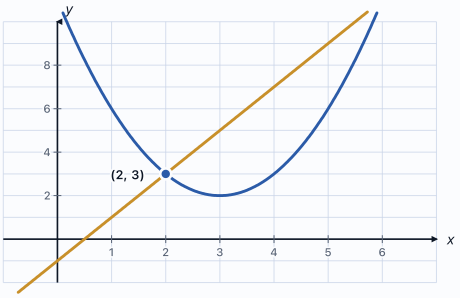
B 3

C 2

D 4

12 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

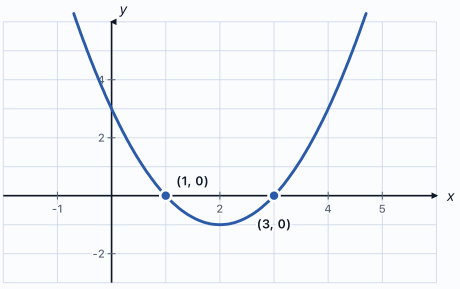
B 4

C 5

D 6

13 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.



A 3

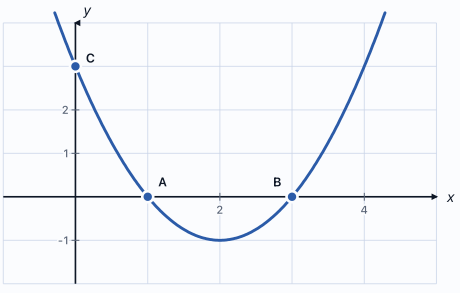
B 4

C -4

D 1

14 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.



A -4

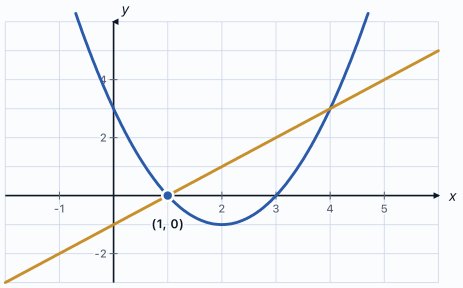
B 4

C 3

D 1

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.



A 1

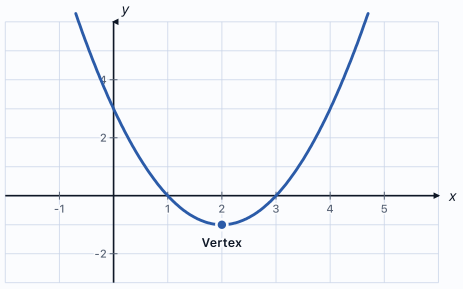
B 8

C 0

D 18

16 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.



A 2

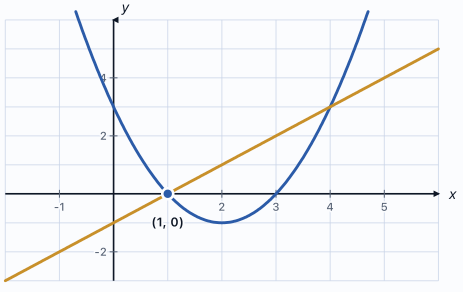
B -2

C 1

D 3

17 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



A (3, 2)

B (5, 4)

C (2, 1)

D (4, 3)

18 MCQ

$\$1200$ is invested at a simple interest rate of 8% per year for 5 years. Find the interest earned.

A 480.00

B 470.00

C 490.00

D 500.00

19 MCQ

Find the midpoint of $A(-1, -1)$ and $B(-7, 5)$.

A $(-6, 6)$

B $(-3, 2)$

C $(-4, 3)$

D $(-4, 2)$

20 MCQ

Find the midpoint of $A(1, 2)$ and $B(5, 10)$.

A $(3, 6)$

B $(3, 7)$

C $(4, 8)$

D $(4, 6)$

21 MCQ

$\$500$ is invested at a simple interest rate of 6% per year for 5 years. Find the interest earned.

A 160.00

B 150.00

C 170.00

D 140.00

22 MCQ

Calculate the midpoint of $A(8, 7)$ and $B(2, 13)$.

A $(-6, 6)$

B $(6, 10)$

C $(5, 11)$

D $(5, 10)$

23 MCQ

Find the midpoint of $A(1, -6)$ and $B(-9, 2)$.

A $(-4, -1)$

B $(-4, -2)$

C $(-10, 8)$

D $(-3, -2)$

24 MCQ

Find the midpoint of $A(-3, -8)$ and $B(-7, -10)$.

A $(-4, -9)$

B $(-4, -2)$

C $(-5, -8)$

D $(-5, -9)$

25 MCQ

$\$2000$ is invested at a simple interest rate of 2% per year for 3 years. Find the interest earned.

A 110.00

B 140.00

C 120.00

D 130.00